12/9/25

Arithmetic Coding:

* we find Tag value.

Encoding → 'DAD'

Formulae:

diff = UL − LL

Range(sym) = LL + diff (P(

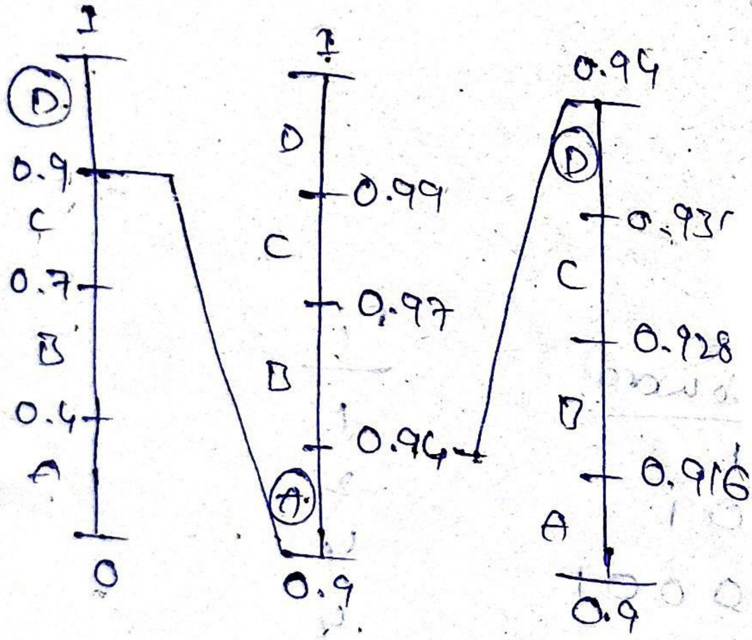
$P(A) = 0.4 \quad [0, 0.4)$

$P(B) = 0.3 \quad [0.4, 0.7)$

$P(C) = 0.2 \quad (0.7, 0.9)$

$P(D) = 0.1 \quad (0.9, 1)$

Upper. Range (sym) = LL + diff (P (sym))



R(A) = 0.9 + 0.1 (0.4)
= 0.94

R (B) = 0.94 + 0.1(0.3)
= 0.97

R(C) = 0.97 + 0.1 (0.2)
= 0.99

D = [0.936, 0.94)

∴ Tag value = $\frac{0.936 + 0.94}{2}$ = 0.938

Q) Find the tag value for the sequence transmitted with the probability distribution

$p(a) = 0.15$

$p(b) = 0.25$

$p(c) = 0.2$

$p(d) = 0.3$

$p(e) = 0.1$

Encode the data, "dabae" using arithmetic coding.

Sol:

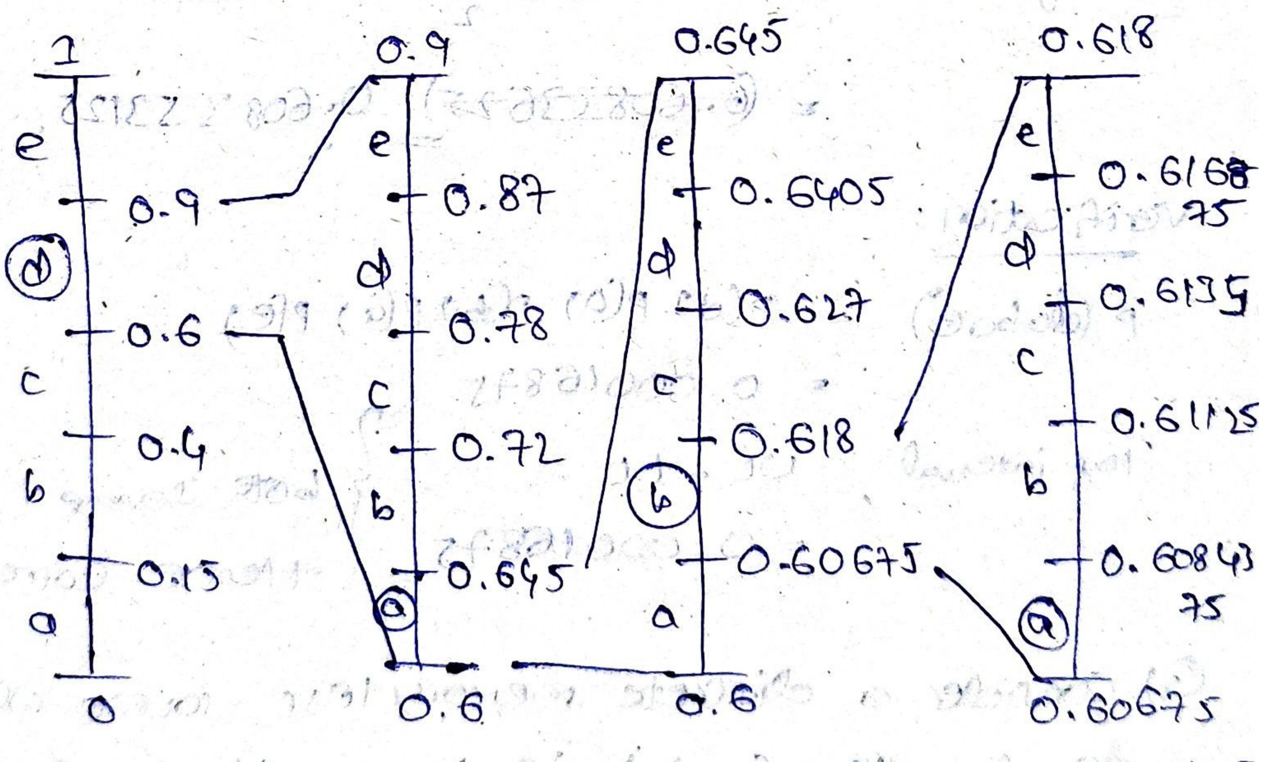
$p(a) = 0.15 \quad [0, 0.15)$

$p(b) = 0.25 \quad [0.15, 0.4)$

$p(c) = 0.2 \quad [0.4, 0.6)$

$p(d) = 0.3 \quad [0.6, 0.9)$

$p(e) = 0.1 \quad [0.9, 1)$



diff = 0.3 $\quad$ diff = 0.045 $\quad$ diff = 0.01125

diff = UL − LL

Upper Range (sym) = LL + diff × p(sym)

0.6084375

e

0.6082~~0~~6875

d

0.607~~7~~625

c

0.607425

b

~~0.6069975~~

0.607003125

a

0.60675

diff = 0.0016875   diff = 0.00020625

Tag interval, e = [0.6082~~0~~6875, 0.6084375)

$\therefore$ Tag value = $\dfrac{0.6082~6875 + 0.6084375}{2}$

= 0.608353125

<u>Verification:</u>

P(dabae) = P(d) P(a) P(b) P(a) P(e)

= 0.00016875

Tag interval = UL − LL

= 0.00016875

both same. Hence, correct.

Q) Consider a discrete memoryless source which decodes the symbol if the received data is 0011101111100011 for the symbols

a	e	i	o	u	!
0.2	0.3	0.1	0.2	0.1	0.1

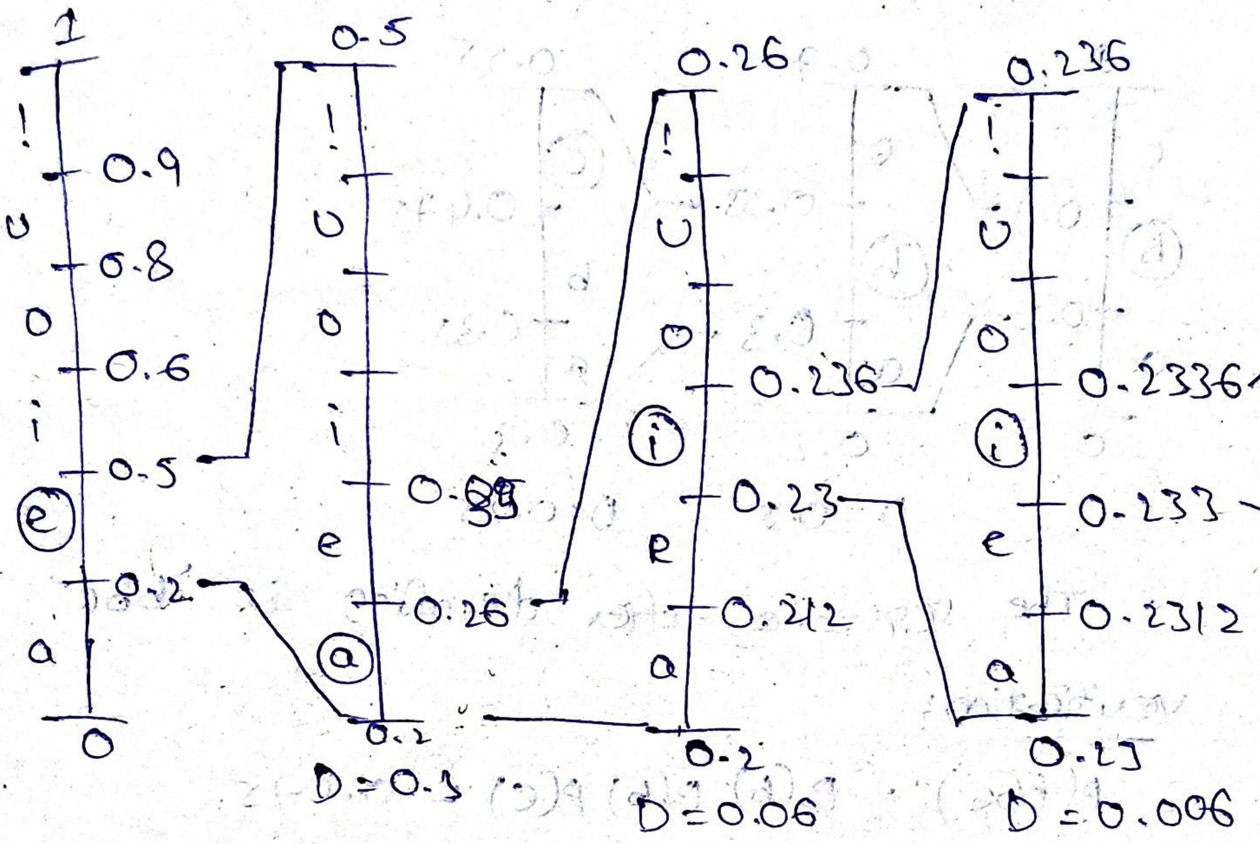
Consider the ! mark as the end of the sequence.

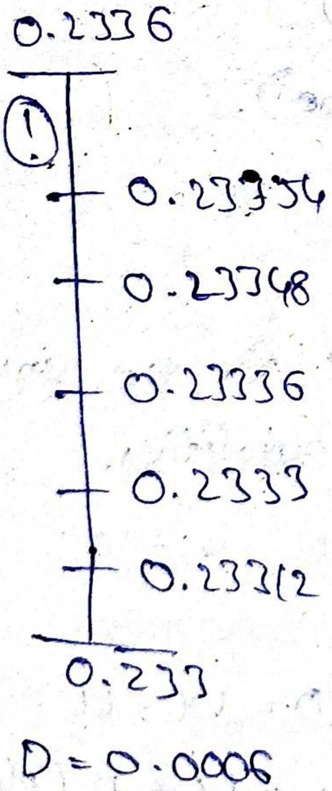
Sol: Consider always, — Arithmetic Decoding

0. 00111 0 11111 0011

$= \frac{1}{2^{1}} \times 0 + \frac{1}{2^{2}} \times 0 + \ldots \ldots = 2^{-1} \times 0 + 2^{-2} \times 0 + 2^{-3} \times 1 + \ldots$

Tag value = 0.23358154З





∴ The sequence after decoding is

"eaii!"

<u>Verification:</u>

P(eaii!) = 0.00006

Tag interval = [0.23354, 0.2336)

d = 0.00006

Hence, correct.

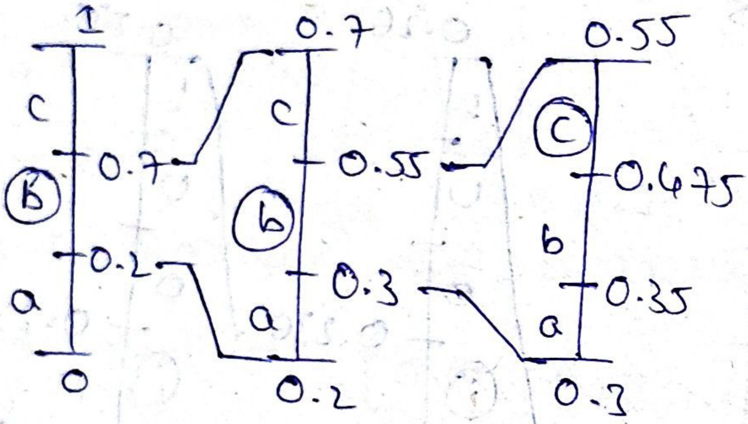
Q) Decode the received codeword ; ... symbols using arithmetic coding (write) probabilities $(a, b, c) = \{0.2, 0.5, 0.3\}$.

Sol: Consider 0.1

$2^{-1} = 0.5$

Tag value $= 0.5$

$$\begin{cases} D = UL - LL \\ U.R\,(sym) = LL + D\,(P(s_i)) \end{cases}$$



$D = 0.5$ $\quad D = 0.25$

∴ The sequence after decoding is "bbc".

<u>Verification:</u>

$P(bbc) = P(b)\,P(b)\,P(c) = 0.075$

Tag interval $= [0.475, 0.55)$

$D = 0.075$ (same)

Hence correct.

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Q) Determine the binary tag value for the sequence of symbols "@#&*" for probabilities.

*	#	@	&
0.3	0.3	0.25	0.15

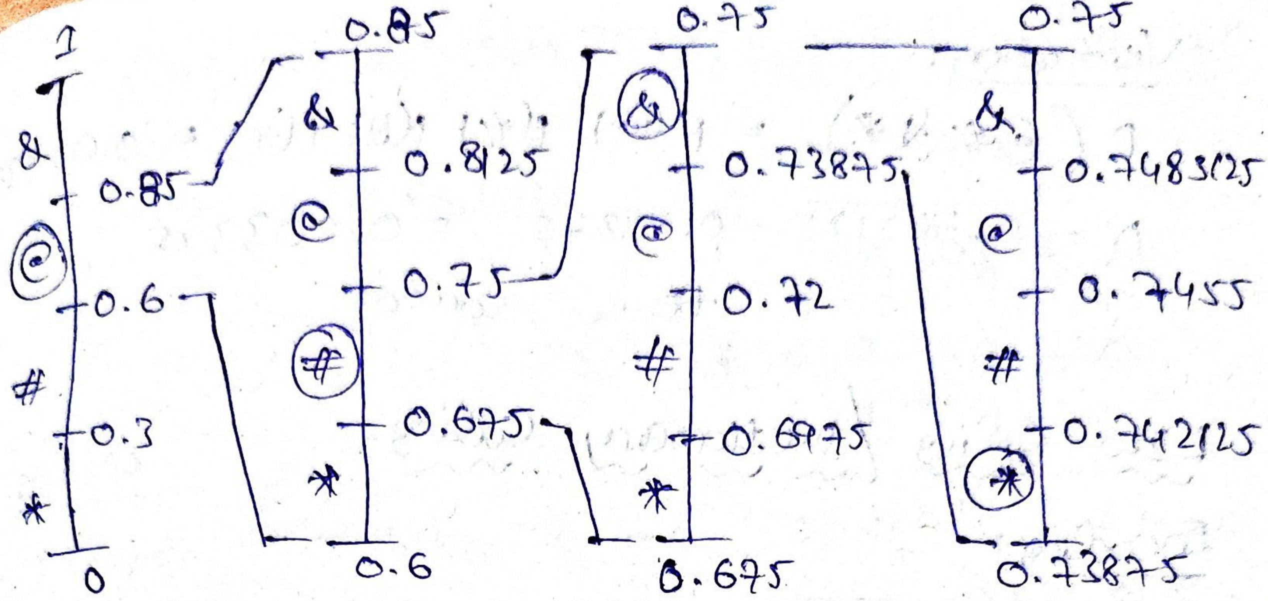
Sol:

$* - [0, 0.3)$

$\# - [0.3, 0.6)$ $\qquad D = UL - LL$

$@ - [0.6, 0.85)$ $\quad U.R\,(s_i) = LL + D\,(P(sym))$

$\& - [0.85, 1)$



D = 0.25 D = 0.075 D = 0.01125

Tag interval = [0.73875, 0.742125)

∴ Tag value = 0.7404375.

Binary Form:

UL		LL
0.742125		0.73875
×2	1 + 0.48425	×2 1 + 0.4775
	0 + 0.9685	0 + 0.955
	1 + 0.937	1 + 0.91
	1 + 0.874	1 + 0.82
	1 + 0.748	1 + 0.64
	1 + 0.496	1 + 0.28
	0 + 0.992	0 + 0.56
	1 + 0.984	1 + 0.12
	1 + 0.968	0 + 0.24

∴ Binary bit is different, so we stop the process.

∴ Binary tag value = "101111011"

Verification:

$P(@\#\&*) = P(@)\,P(\#)\,P(\&)\,P(*) = 0.003375$

$D = 0.742125 - 0.73875 = 0.003375$

Hence, correct

LZW Coding / Dictionary Coding:

Encoding:

sequence data : ω a b b a ƀ ω a b b a
ƀ ω a b b a ƀ ω a b b
a ƀ ω

i/p = 25 Entries = 5

Index	Entry	Index	Entry	Index	Entry
1	ƀ	6	ωa (5,a)	11	ƀω
2	a	7	ab	12	ωab
3	b	8	bb	13	bba
4	o	9	ba	14	aƀω
5	ω	10	aƀ	15	ωabb

Index	Entry
16	baƀ
17	ƀωa
18	abb
19	baƀω

o/p = 12

Entries = 19.

5 2 3 9 1 6 8 10 12 16
ω | a | b | b a | ƀ | ω a | b b | a ƀ | ω a b | b a

15 14
5 | ω ...

Encoded seq = 5, 2, 3, 9, 1, 6, 8, 10, 12, 16, 15, 14

Compression Ratio $CR = \dfrac{25 \log_2 25}{12 \log_2 19}$

$\therefore$ CR = 1.349344

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Q) Encode the following msg abaababababab using LZW coding where a = 0 and b = 1.

Sol: Given, A B A AB A B A B A B A B.

i/p = 13 Entries = 2

Index	Entry
0	A
1	B
2	AB
3	BA

Index	Entry
4	AA
5	ABA
6	~~ABAB~~
7	BAB

0	1	4	3	7	2	2
A	B	A A	B A	B A B	A B	A B

$\therefore CR = \dfrac{13 \log_2 (2)}{7 \log_2 (7)} = 0.66152$

Q) Determine the CR for the image patch given below if it undergoes LZW coding. Consider the initial dictionary as 0 - 255

$$I = \begin{bmatrix} 65 & 65 & 68 & 67 \\ 67 & 65 & 65 & 67 \\ 67 & 65 & 67 & 67 \\ 67 & 65 & 67 & 67 \end{bmatrix}$$

Sol: Make I as one-dimensional array.

65 65 68 67 . 63 65 65 67 63 65 67 67
63 65 67 67

$i/p = N_1 = 16$

Entries $= 256 = a_1$

Index	Entry
0	0
⋮	⋮
63	63
65	65
67	67
68	68
⋮	⋮
255	255
256	65 65
257	65 68

Index	Entry
258	68 67
259	67 63
260	63 65
261	65 65 67
262	67 63 65
263	65 67
264	67 67
265	67 63 65 67

~~O/P = N_2~~

65	65	68	67	63	256	259	263	262	264
65	65	68	67	63	65 65	67 63	65 67	67 63 65	67 67

$O/P = N_2 = 10$, Entries $= a_2 = 266$

$$\therefore CR = \frac{16 \log_2 256}{10 \log_2 266} = 1.589019$$

Decoding : (LZW)

{314 684 212 5 10 6 11 13 6}

sequence = 3 1 4 6 8 4 2 1 2 5 10 6 11 13 6

3	1	4	6	8	4	2	1	2	5	10	6	11	13	6
r	a	t	at	ata	t	b̸	a	b̸	ra	tb̸	at	b̸a	b̸r	at

Index	Entry
1	a
2	b̸
3	r
4	t
5	ra
6	at
7	ta
8	ata
9	atat
10	tb̸
11	b̸a
12	ab̸
13	b̸r

$N_1 = 15 \quad a_1 = 4$

$N_2 = 24 \quad a_2 = 13$

$$\therefore CR = \frac{15 \log_2 4}{24 \log_2 13} = 0.337797$$

19/09/25,

Q) Encode the text data wys_wygwys_wy swysg. Consider the initial dictionary coding as ascii values 0-255.

Sol: Given,

wys_wygwys_wyswysg

$N_1 = 18 \quad a_1 = 256$

A - 65
G - 71
S - 83
W - 87
Y - 89
Z - 90

Index	Entry
0	0
⋮	
65	A
71	G
83	S
87	W
89	Y
90	Z
95	_
⋮	⋮
255	255

Index	Entry	Index	Entry
256	WY	262	WYS
257	YS	263	S_W
258	S_	264	WYSW
259	_W	265	WYSG
260	WYG	~~266~~	
261	GW		

87	89	83	95	256	71	262	259	257	265
W	Y	S	_	W Y	G	W Y S	_ W	Y S	W Y S G

$N_2 = 10 \qquad a_2 = 266$

$$\therefore CR = \frac{N_1 \log_2(a_1)}{N_2 \log_2(a_2)} = \frac{18 \log_2(256)}{10 \log_2(266)} = 1.20808$$

Q) For the coded sequence $c_q = 0102537$ and the initial dictionary is $d = \{(A,0),(0,1)\}$ produced the original message using LZW coding.

Sol: $C_d = 0\ 1\ 0\ 2\ 5\ 3\ 7$ $d = \{(A,0), (B,1)\}$

0	1	0	2	5	3	7
A	B	A	AB	ABA	BA	BAB

Index	Entry
0	A
1	B
2	AB
3	BA
4	AA
5	ABB

Index	Entry
6	ABAB
7	BAB

$N_1 = 7$ $a_1 = 2$

$N_2 = 13$ $a_2 = 7$

O. seq = A B A AB ABA BA BAB.

$$\therefore \ CR = \frac{7 \log_2(2)}{13 \log_2(7)} = 0.179487$$

Q) Perform LZW coding and determine the sequence 2 1 3 2 4 6 5 2 4 8 10 3 13. Consider the initial dictionary as

$d = \{(@,1), (\#,2), (\$,3), (☆,4)\}$

Sol:-

2	1	3	2	4	6	5	2	4	8	10	3	13
#	@	$	#	☆	@$	#@	#	☆	#☆	@$#	$	#☆@

Index	Entry
1	@
2	#
3	$
4	☆

Index	Entry
5	#@
6	@$
7	$#
8	#☆

Index	Entry
9	☆@
10	@$#
11	#@#
12	#☆#
13	#☆@

$N_1 = 13 \quad q_1 = 4$

$N_2 = 20 \quad q_2 = 13$

O.Sq :

@ $ # ☆ @$ # @ #☆# ☆ @$#$ #☆@

$$\therefore \quad CR = \frac{13 \log_2(4)}{20 \log_2(13)} = 0.351309$$